AI 101 for Mechanical Engineers · Module 7

Three useful metrics and their limits

Lesson 7.2

What you will learn

> Calculate MAE and RMSE, interpret R^2, and explain why aggregate metrics should be accompanied by residual inspection.

Lesson overview

Module 7 · Lesson 2

Three useful metrics and their limits

AI 101 for Mechanical Engineers

1 Start with signed residuals

$$r_i = y_i - \hat{y}_i$$

y_i: measured output · $\hat{y}_i$: prediction · i: observation index

Measured (K)	Predicted (K)	Residual (K)
5	4	+1
8	10	−2
10	7	+3

Positive residual: underprediction.
Negative residual: overprediction.

2 MAE and RMSE keep target units

Mean absolute error (MAE)

$$\mathrm{MAE} = (1/n)\,\Sigma\,|r_i|$$

For these data: $(1 + 2 + 3)\ \mathrm{K} / 3 = 2\ \mathrm{K}$

Root mean squared error (RMSE)

$$\mathrm{RMSE} = \sqrt{[\ (1/n)\,\Sigma\ r_i{}^2\ ]}$$

For these data: $\sqrt{(14\ \mathrm{K}^2 / 3)} \approx 2.160\ \mathrm{K}$

Σ sums over all n evaluated observations (n is a dimensionless count).

RMSE gives larger errors more influence. RMSE ≥ MAE for the same errors; they are equal only when all error sizes are equal.

3 R^2 is a relative score

$$R^2 = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}}$$

SSE: sum of squared prediction residuals.
SST: sum of squared deviations of the targets from their evaluation-set mean.

The squared units cancel, so R^2 is dimensionless.

For these data: $R^2 \approx -0.1053$

A negative R^2 means a larger squared error than simply predicting the evaluation-set mean.

If the targets are constant, SST = 0 and the formula needs explicit handling.

Key takeaway

Inspect signed residuals across load, time and operating regimes. One overall score can hide large local errors.
A high R^2 does not prove causation, small physical error or safe operation.

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Signed residuals show under- and overprediction

$$r_i = y_i - \hat{y}_i$$

- Positive: underprediction; negative: overprediction
- Measured [5, 8, 10] K; predicted [4, 10, 7] K
- Residuals: [1, −2, 3] K

MAE and RMSE keep the target unit

$$\mathrm{MAE} = (1/n)\,\Sigma\,|r_i|$$

$$\mathrm{RMSE} = \sqrt{((1/n)\,\Sigma\,r_i^2)}$$

- n: number of evaluated observations, a dimensionless count
- MAE removes the signs before averaging
- RMSE squares errors, giving larger magnitudes more influence

Worked example: MAE 2 K, RMSE ≈ 2.160 K

$$\mathrm{MAE} = (1 + 2 + 3)\ \mathrm{K} / 3 = 2\ \mathrm{K}$$
$$\mathrm{RMSE} = \sqrt{((1 + 4 + 9)\ \mathrm{K}^2 / 3)} \approx 2.160\ \mathrm{K}$$

- RMSE is at least as large as MAE for the same errors [35]
- They are equal when all absolute residuals are equal

R^2 compares the model with mean prediction

$$R^2 = 1 - SSE/SST$$

- R^2 formula as given in [16, Sec. 3.1.3]
- SSE: sum of squared prediction residuals
- SST: sum of squared target deviations from evaluation-set mean
- Squared target units cancel; R^2 is dimensionless

R^2 can be negative or undefined

- Negative when model squared error exceeds the mean comparison [91]
- The example gives approximately -0.1053
- Constant target: SST is zero and the formula undefined
- scikit-learn r2_score default: 1.0 if perfect, otherwise 0.0 [91]

State which rule you use. High R^2 does not establish causation or safe operation.

Neither MAE nor RMSE is universally best

MAE

- Averages absolute errors

RMSE

- Squares, averages, then takes a square root
- Gives large residuals more influence

> Compare with the cost of large errors; inspect residuals by operating condition.

Inspect where errors occur

- Plot signed residuals against load, time and operating regime
- Keep the residuals in physical units
- An aggregate score can hide a narrow region of severe errors
- A residual plot can reveal hidden patterns [37]

Residual patterns are clues, not unique explanations

Curving with load

- Straight line may miss structure

Step change

- May point to another regime or omitted input

Check data quality, operating context and model assumptions before choosing a correction.

CHECK YOUR UNDERSTANDING

For residuals −2 K and +2 K, are MAE and RMSE equal?

Answer before moving on.

CHECK YOUR UNDERSTANDING

For residuals −2 K and +2 K, are MAE and RMSE equal?

> Yes. Both are 2 K because the absolute magnitudes are equal. Averaging signed residuals to get zero would not calculate either metric.

Key takeaways

- Preserve residual signs for interpretation
- MAE and RMSE report error in target units
- R^2 is dimensionless, can be negative, needs care for constant targets
- Inspect errors across operating conditions, not only the aggregate score

References

[16] G. James, D. Witten, T. Hastie, and R. Tibshirani, An Introduction to Statistical Learning: With Applications in R, 2nd ed. New York, NY, USA: Springer, 2021. doi: 10.1007/978-1-0716-1418-1

[35] C. J. Willmott and K. Matsuura, “Advantages of the mean absolute error (MAE) over the root mean square error (RMSE) in assessing average model performance,” Clim. Res., vol. 30, pp. 79–82, 2005. doi: 10.3354/cr030079

[37] F. J. Anscombe, “Graphs in statistical analysis,” Amer. Statistician, vol. 27, no. 1, pp. 17–21, 1973. doi: 10.1080/00031305.1973.10478966

[91] scikit-learn developers, scikit-learn User Guide and API Reference, ver. 1.9.1. [Online]. Available: https://scikit-learn.org/stable/ Accessed: Oct. 20, 2026.

Numbers match the course reference list. The written lesson is the authoritative text.